

HopGraph

Find the cheapest cross-chain path with real-time graph search, not guesswork

The Problem & Our Solution

Cross-chain transfers hide fees across dozens of bridges and DEXs, so users and LPs overpay without knowing it.

- Bridges and DEXs have opaque, inconsistent fee structures across chains
- Manually comparing routes is slow, confusing, and error-prone
- HopGraph models the multi-chain landscape as a weighted graph
- Dijkstra/A* search finds the lowest-total-cost path in real time
- One interface replaces manual bridge-hopping and guesswork



How It Works

Users pick source/destination assets; HopGraph searches live fee data and executes the cheapest multi-hop route.

- User selects source chain, destination chain, and asset to move
- Engine pulls live fee, slippage, gas, and time data per edge
- Dijkstra/A* ranks top-3 cheapest paths across bridges and DEXs
- User picks a path; execution runs via Wormhole, deBridge, Jupiter SDKs
- Dashboard shows savings versus the naive single-bridge route

Built with

Solana, Ethereum, Arbitrum, Base,
TypeScript, Rust, Dijkstra/A* pathfinding

Why Now & What's Next

Liquidity is fragmented across dozens of chains, so manual comparison no longer scales and users overpay daily.

- Bridges and L2s have multiplied, fragmenting liquidity and fees
- Next: add more bridges/DEXs with oracle-based live weight refresh
- Next: LP rebalancing mode with impermanent loss modeling
- Next: public API/widget so wallets and dApps can embed HopGraph

Category

DeFi